

47 Given Design Requirements

I will choose the minimum allowed load:

$$P = 20 \text{ kN}$$

dimension:

$$a = 0.4 \text{ m}$$

$$b = 0.3 \text{ m}$$

supports are:

$$A = \text{pin}$$

$$B = \text{roller}$$

Loads are:

$$C = 20 \text{ kN upward}$$

$$D = 20 \text{ kN downward}$$

$$SF_{\text{truss}} = 3.5$$

$$SF_{\text{pin}} = 4$$

$$\tau_y = 170 \text{ ksi}$$

$$\sigma_{\text{pin}} = 0.298 \text{ lb/in}^3$$

Material selection: (solid works library)

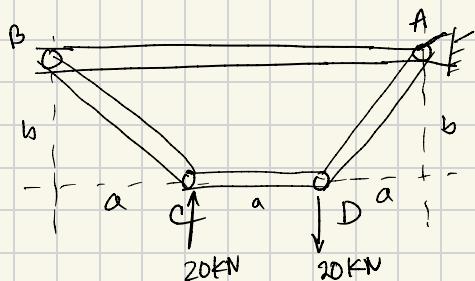
AISI 1020

$$\text{Yield Strength } S_y = 351.571 \text{ MPa}$$

$$\rho = 7900 \text{ kg/m}^3$$

$$\text{Elastic Modulus} = 200 \text{ GPa}$$

b) Truss Geometry?



BC, BD, CD, CA, DA

$$m = 2j - 3$$

$$m = 5$$

$$j = 4$$

$$2(4) - 3 = 5$$

$$m = 5$$

length:

$$BC = \sqrt{a^2 + b^2} = 0.5 \text{ m} \quad CD = a = 0.4 \text{ m}$$

$$DA = \sqrt{a^2 + b^2} = 0.5 \text{ m}$$

$$BD = \sqrt{(2a)^2 + b^2} = 0.854 \text{ m}$$

$$CA = 0.854 \text{ m} = BD$$

⑦ Supported Reaction:

$$\sum M_B = 0:$$

$$A_y(3a) + P(a) - P(2a) = 0$$

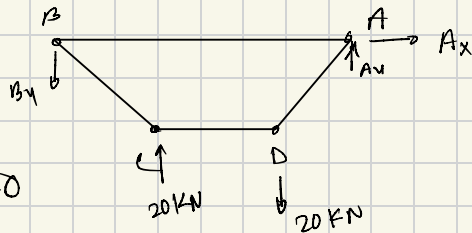
$$A_y = P/3 = 20/3 = 6.67 \text{ kN (Upward)}$$

$$\sum F_y = 0$$

$$B_y + A_y + P - P = 0$$

$$B_y = -6.67 \text{ kN (downward)}$$

$$\sum F_x = 0; A_x = 0$$



⑧ Member Force Analysis

Joint C:

$$\sum F_x = 0: -F_{BC} \cos \theta + F_{CD} + F_{CA} \cos \phi = 0$$

$$\sum F_y = 0: -F_{BC} \sin \theta + F_{CA} \sin \phi + 20 = 0$$

Joint D:

$$\sum F_x = 0: F_{DA} \cos \theta - F_{CD} - F_{BD} \cos \phi = 0$$

$$\sum F_y = 0: F_{DA} \sin \theta - F_{BD} \sin \phi - 20 = 0$$

Joint A

$$\sum F_x = 0 : -F_{DA} \cos \theta - F_{CA} \cos \phi = 0$$

$$\sum F_y = 0 : +F_{DA} \sin \theta - F_{CA} \sin \phi + 6.67 = 0$$

Joint B

$$\sum F_x = 0 : F_{BC} \cos \theta - F_{BD} \cos \phi = 0$$

$$\sum F_y = 0 : F_{BC} \sin \theta + F_{BD} \sin \phi - 6.67 = 0$$

For Joint B:

$$F_{BC}(0.8) = F_{BD}(0.936) \Rightarrow F_{BC} = 1.17 F_{BD}$$

$$1.17 F_{BD}(0.6) + F_{BD}(0.351) = 6.67$$

$$0.702 F_{BD} + 0.351 F_{BD} = 6.67$$

$$F_{BD} = 6.33 \text{ kN (C)}$$

$$F_{BC} = 7.41 \text{ kN (C)}$$

From Joint A:

$$F_{DA}(0.8) + F_{CA}(0.936) = 0 \Rightarrow F_{DA} = -1.17 F_{CA}$$

$$F_{DA}(0.6) + F_{CA}(0.351) = 6.67$$

$$-1.17 F_{CA}(0.6) + 0.351 F_{CA} = 6.67$$

$$F_{CA} = -19.00 \text{ kN (C)}$$

$$F_{DA} = +1.17 \times (-19.00) = 22.23 \text{ kN (tension)}$$

$$F_{CD} = 0$$

Large force $F_{DA} = 22.23 \text{ kN}$ (Tension)

$$F_{\max} = F_{DA}$$

4) Truss Member cross-section

$$F_{\max} = 22.23 \text{ kN}, \quad SF = 3.5$$

$$S_y = 351.521 \times 10^6 \text{ Pa}$$

$$A_{\min} = \frac{F_{\max} \cdot SF}{S_y} = 221.3 \text{ mm}^2$$

Selected cross section

$(15 \times 15) \text{ mm}^2$ bar = 225 mm^2 slightly larger than min.

4) Approximate Weight of Truss Members

$$V = 225 \times 10^{-6} \times 3.108 = 6.993 \times 10^{-4} \text{ m}^3$$

$$m = 6.993 \times 10^{-4} \times 7900 = 5.52 \text{ kg}$$

$$W_{\text{truss}} = 5.52 \times 9.81 = 54.2 \text{ N}$$

4) Pin Design:

Largest pin force = 6.62 kN

$$A_{\text{pin}} = \frac{4 \cdot V_{\max}}{\tau_y}$$

$$A_{\text{pin}} = \frac{4 \times 6670}{1.1721 \times 10^8} = 22.75 \text{ mm}^2$$

Selected pin diameter = 6 mm